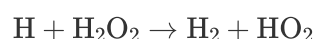
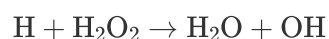
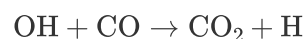
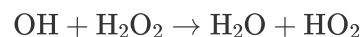
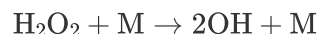


Question

The experimenters attempted to investigate the decomposition mechanism of hydrogen peroxide. When carbon monoxide is introduced into the system, the decomposition mechanism of hydrogen peroxide is considered to be as follows:



(In fact, there is also the reaction $\text{OH} + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{H}$, but it is known that there is no H_2 at the initial stage of the reaction, and the rate of H producing H_2 is much lower than the rate of producing H_2O (about 350 times), so this elementary reaction is not considered in this question.)

The rate constants of these reactions are $k_1 \sim k_6$ ($[\text{M}]$ has been included in k_1 , and $[\text{M}]$ does not need to be involved in the calculation). Derive the rate equation for the decomposition of H_2O_2 using the steady-state approximation method, and express it in terms of $[\text{H}_2\text{O}_2]$, $[\text{CO}]$, k_1 , k_2 , k_4 , k_5 , k_6 . The following statements are made:

1. The rate equation for the decomposition of H_2O_2 is $r = 2k_1[\text{H}_2\text{O}_2] \left[\frac{1 + \frac{k_2[\text{H}_2\text{O}_2]}{k_4[\text{CO}]}}{\frac{k_6}{k_5 + k_6} + k_2[\text{H}_2\text{O}_2]} \right]$
2. When $[\text{CO}]$ is very small or $[\text{CO}]$ is very large, the order of the reaction with respect to H_2O_2 is 1
3. A mixed gas of $\text{CO} \cdot \text{N}_2$ is used to determine the rate constant. Let r_{N} be the reaction rate when $[\text{CO}] = 0$, and r_{CO} be the reaction rate when $[\text{CO}] \neq 0$. At a certain temperature, the measured values of $r_{\text{N}}/r_{\text{CO}}$ and $[\text{H}_2\text{O}_2]/[\text{CO}]$ are shown below:

r_N/r_{CO}	0.153	0.217	0.318	0.474	0.516
$[H_2O_2]/[CO]$	0.00192	0.00681	0.0167	0.0384	0.0472

Then it can be calculated that $k_2/k_4 = 17.2$, $k_5/k_6 = 7.02$

4. After a series of experiments, researchers believe that the addition of the following elementary reaction can better describe the decomposition of hydrogen peroxide: $HO_2 + CO \rightarrow CO_2 + OH$

Let $[HO_2] = a_1$, $r = b_1$ when this reaction is not considered, and $[HO_2] = a_2$, $r = b_2$ when this reaction is considered. Keeping $[H_2O_2] \cdot [CO]$ constant, it can be concluded from the steady-state approximation method that $a_1 = a_2$, $b_1 > b_2$

The following options contain all the correct statements:

A. All other options are incorrect

B. 1,2

C. 1

D. 2,3

E. 2,4

F. 1,3,4

G. 1,2,4

H. 3

I. 2,3,4

J. 1,2,3,4

K. 1,4

L. 3,4

Answer

Correct Answer: **D**

Detailed Explanation

Perform steady-state approximation for the three intermediates OH , HO_2 , H :

$$d[\text{OH}]/dt = 2k_1[\text{H}_2\text{O}_2] - k_2[\text{OH}][\text{H}_2\text{O}_2] + k_5[\text{H}][\text{H}_2\text{O}_2] - k_4[\text{OH}][\text{CO}] = 0 \quad (1)$$

CHECKPOINT

1 PTS

$$d[\text{OH}]/dt = 2k_1[\text{H}_2\text{O}_2] - k_2[\text{OH}][\text{H}_2\text{O}_2] + k_5[\text{H}][\text{H}_2\text{O}_2] - k_4[\text{OH}][\text{CO}] = 0$$

$$d[\text{HO}_2]/dt = k_2[\text{OH}][\text{H}_2\text{O}_2] + k_6[\text{H}][\text{H}_2\text{O}_2] - 2k_3[\text{HO}_2]^2 = 0 \quad (2)$$

CHECKPOINT

1 PTS

$$d[\text{HO}_2]/dt = k_2[\text{OH}][\text{H}_2\text{O}_2] + k_6[\text{H}][\text{H}_2\text{O}_2] - 2k_3[\text{HO}_2]^2 = 0$$

$$d[\text{H}]/dt = k_4[\text{OH}][\text{CO}] - k_5[\text{H}][\text{H}_2\text{O}_2] - k_6[\text{H}][\text{H}_2\text{O}_2] = 0 \quad (3)$$

CHECKPOINT

1 PTS

$$d[\text{H}]/dt = k_4[\text{OH}][\text{CO}] - k_5[\text{H}][\text{H}_2\text{O}_2] - k_6[\text{H}][\text{H}_2\text{O}_2] = 0$$

$$(1) + (3): k_2[\text{OH}] + k_6[\text{H}] = 2k_1, \quad [\text{OH}] = (2k_1 - k_6[\text{H}])/k_2 \quad (4)$$

Substituting (4) into (3): $[H] = \frac{2k_1k_4[CO]}{k_4k_6[CO] + k_2(k_5 + k_6)[H_2O_2]}$

And $r = k_1[H_2O_2] + k_2[OH][H_2O_2] - k_3[HO_2]^2 + k_5[H][H_2O_2] + k_6[H][H_2O_2] = 2k_1[H_2O_2] + k_5[H][H_2O_2]$ (5)

Thus $r = 2k_1[H_2O_2] \left[\frac{1 + \frac{k_2[H_2O_2]}{k_4[CO]}}{\frac{k_6}{k_5 + k_6} + \frac{k_2[H_2O_2]}{k_4[CO]}} \right]$

CHECKPOINT

3 PTS

$$r = 2k_1[H_2O_2] \left[\frac{1 + \frac{k_2[H_2O_2]}{k_4[CO]}}{\frac{k_6}{k_5 + k_6} + \frac{k_2[H_2O_2]}{k_4[CO]}} \right]$$

Statement 1: The denominator omits $k_4[CO]$, incorrect.

CHECKPOINT

1 PTS

Statement 1 is incorrect

Statement 2: When $[CO]$ concentration is very small, $\frac{k_2[H_2O_2]}{k_4[CO]} \gg 1$, at this time $r \approx 2k_1[H_2O_2]$; when $[CO]$ concentration is very large, $\frac{k_2[H_2O_2]}{k_4[CO]} \ll 1$, at this time $r \approx 2k_1[H_2O_2] \frac{k_5 + k_6}{k_6}$. For $[H_2O_2]$ all are first order, Statement 2 is correct.

CHECKPOINT

2 PTS

When $[CO]$ is very large or very small, for $[H_2O_2]$ all are first order, Statement (2) is correct

Statement 3: Transformation yields:

$$\frac{r_N}{r_{CO}} = \frac{k_6}{k_5 + k_6} + \frac{k_2[H_2O_2]}{k_4[CO]} \left(1 - \frac{r_N}{r_{CO}} \right)$$

CHECKPOINT

2 PTS

$$\frac{r_N}{r_{CO}} = \frac{k_6}{k_5+k_6} + \frac{k_2[H_2O_2]}{k_4[CO]} \left(1 - \frac{r_N}{r_{CO}}\right)$$

Substituting numerical values, for linear regression of $\frac{[H_2O_2]}{[CO]} \left(1 - \frac{r_N}{r_{CO}}\right) - \frac{r_N}{r_{CO}}$: $k_2/k_4 = 17.2$, $k_5/k_6 = 7.02$, Statement 3 is correct.

CHECKPOINT

2 PTS

$k_2/k_4 = 17.2$, $k_5/k_6 = 7.02$, Statement 3 is correct

Statement 4: From equation (1)+(2)+(3), it can be known that $2k_1[H_2O_2] = k_3[HO_2]^2$, considering that the seventh reaction does not affect this result, thus $a_1 = a_2$

CHECKPOINT

1 PTS

$$a_1 = a_2$$

Similarly, rewriting the expression for r can reveal that $r = [OH](k_2[H_2O_2] + k_4[CO])$, adding a reaction that generates OH leads to an increase in its concentration, thus $r_2 > r_1$

CHECKPOINT

1 PTS

$$r_2 > r_1$$

Statement 4 is incorrect.

Therefore, choose D.

$$\text{In fact, } r = 2k_1[\text{H}_2\text{O}_2]\left[1 + \frac{k_7[\text{CO}]}{\sqrt{k_1k_3[\text{H}_2\text{O}_2]}}\right]\left[\frac{1 + \frac{k_2[\text{H}_2\text{O}_2]}{k_4[\text{CO}]}}{\frac{k_6}{k_5+k_6} + \frac{k_2[\text{H}_2\text{O}_2]}{k_4[\text{CO}]}}\right]$$